Hanoi, November 10, 2025

To: The Admissions Committee
University of Michigan-Dearborn
Dearborn, MI 48128, USA

Letter of Recommendation for Xuan Thanh Trinh

Dear Members of the Admissions Committee,

I am Nguyen Duc Tuyen, an Associate Professor and Head of the Power Grid and Renewable Energy (PGRE) Laboratory at Hanoi University of Science and Technology (HUST). I also serve as an Adjunct Associate Professor at the Shibaura Institute of Technology (Japan), where I earned my PhD in Electrical Engineering. Over the past decade, my research has focused on smart grids, microgrids, and the integration of renewable energy systems, leading to over 100 scientific publications. Having mentored numerous engineering students throughout my academic career across institutions in Japan and Vietnam, I write to you today to strongly recommend my student, Xuan Thanh Trinh, for admission to the PhD program at the University of Michigan-Dearborn.

Among the students I have advised, Thanh stands out as a capable and practical researcher. He joined my research group for two years with a clear interest in using mathematical optimization to address renewable energy integration. From the beginning, he demonstrated a methodical approach to his work. He did not simply apply standard computational tools; he proactively analyzed literature, identified research gaps, and translated mathematical models into functional code. Notably, when encountering complex technical errors that challenged other lab members, he analyzed the issues and refined his approach. He consistently sought to improve his initial results, ensuring his solutions met practical grid requirements. This persistence and dedication to problem solving led directly to a successful oral presentation at the ICGEA conference and an excellent Bachelor's thesis.

Thanh's strong discipline and time management are evident in his ability to balance research, coursework, and industry practice. While actively contributing to our lab, he consistently ranked in the top 3 of his class, demonstrating a solid understanding of Power System principles. He further validated this theoretical knowledge through hands-on experience during his internship at the Northern Electrical Testing Company (ETC1). Successfully managing these responsibilities highlights the capability and strong sense of responsibility he will bring to your PhD program.

Building on these qualities, Thanh has also shown a natural ability to guide and support others within our laboratory. Rather than focusing solely on his own projects, he proactively built automated workflows to assist the entire team. He also readily dedicates his time to supporting new members, patiently guiding them through algorithmic modeling. This collaborative mindset has fostered a positive environment and contributed to the lab's overall productivity.

Considering his academic background, research experience, and practical skills, I believe Thanh is well-prepared for graduate studies. I am pleased to recommend him for admission to the PhD program at the University of Michigan-Dearborn, as I am confident he has the potential to be a valuable addition to your research community and the broader field of power systems engineering.

If you need any further information regarding Thanh, please feel free to contact me via email at tuyen.nguyenduc@hust.edu.vn.

Sincerely,

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